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Deep Learning for Football Outcomes Prediction based on Football Rating System

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Abstract. The prediction models of association football can be categorized into three natural clusters, which are the statistical models, the machine learning and probabilistic graphical models and rating systems. The prediction may focus on matches outcomes prediction (win, draw and lose) or the number of goals scored obtained by home and away team. Match result prediction is very important as a benchmark for the management team to assess the chances of winning, drawing and losing for a particular team before the match starts. In the 2017 soccer challenge, the conventional machine learning and ensemble methods, as well as probabilistic graphical models such as k -Nearest Neighbor (k -NN), Gradient Boosting Tree (GBT) and Bayesian Networks (BN) have been a choice of researchers that participate the challenge and successfully dominate the challenge. Even so, there are some well-known techniques such as Neural Networks (NN) and Deep Neural Networks (DNN) that are left in this challenge although this technique has advantages in producing good results in many fields including solving real world problems which the results are comparable and sometimes superior to expert knowledge in some cases. In this paper, we propose a football match outcome prediction based on pi-rating system using TabNet, a DNN architecture for tabular data. The experiment cover two parts namely: (1) generates pi-rating system from 216,743 instances of raw football dataset and (2) predicts 206 football match outcomes using TabNet. As a result, the proposed prediction model based on the pi-rating system using TabNet successfully overcomes the existing model for football match outcomes for both scoring rules, in terms of accuracy and Rank Probability Score (RPS). The findings from this study are important because they can be used for future researchers in developing new football match outcome prediction models that incorporate several new features along with other features. This is important to improve the predictive performance of football prediction models using other advanced techniques.

INTRODUCTION

Predicting association football matches has been historically a popular research topic either in terms of predicting the numbers of goals scored or predicting the match outcomes (win, draw and lose). In general, the prediction models of association football form three natural clusters, which are the statistical models, the machine learning and probabilistic graphical models, and rating systems [1]. [1] also explained that these groups can be categorized according to certain techniques and published in certain journals as in Figure 1. Nevertheless, the concern in football prediction modelling can be observed circles around three problems of statement which are (1) the football data [2], (2) approaches, algorithms, or techniques [3] and (3) performance metric, scoring rules or evaluation metric [4].

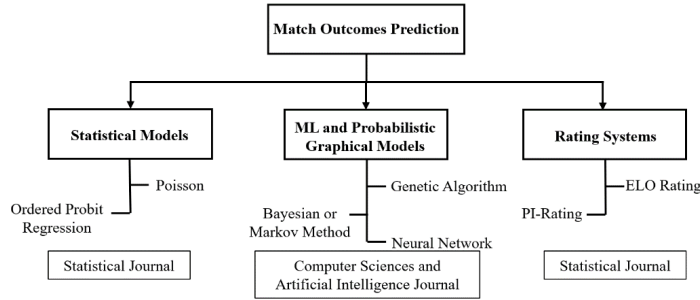


FIGURE 1. A football prediction taxonomy according to Constantinou (2019)

The 2017 Soccer Prediction Challenge can be seen as one of initiative to gather top researchers to solve some problems occurred around football prediction which focus on football data and using variation of approaches, algorithms, or techniques. [2] stated that predictive modelling in football poses challenges with respect to the integration of domain knowledge and feature engineering. It is because predicting football matches outcomes have common restriction in term of availability of sophisticated relevant data that can be used for developing football prediction model. Although there are many freely accessible data on sports websites, its only limited to several basic data features.

The conventional machine learning and ensemble methods as well as probabilistic graphical models such as k -Nearest Neighbor (k -NN), Gradient Boosting Tree (GBT) and Bayesian Networks (BN) have been a choice of researchers that participate the challenge and successfully dominate the challenge. Though, there are some notable techniques such as Neural Networks (NN) and Deep Neural Networks (DNN) with have history in football prediction development like in work of [5], [6] and [7], it still has been left out in this challenge. Each work shows the improvement of application of NN and DNN in football prediction. For example, [5] exclude the prediction of draw results because their models don't not have capability to predict draw meanwhile [6] and [7] work can predict the draw results.

Hence, in this study the prediction performance for football matches outcomes is investigated based on the pi-rating system which generated from the limited acquired dataset using TabNet a new DNN architecture for tabular data. The work of [8] and [9] shows that TabNet obtained best performance compared to traditional DNN and several ensemble methods such as Catboost, Xgboost and LightGBM due to capability of TabNet to incorporate tabular data compare to traditional DDN and at the same time starts to challenges the domination of ensemble method especially boosting tree where the capabilities of transformer in TabNet architecture that usually be applied in natural language and image data now taking place in tabular data where its functionality resembles to tree ensembles. Besides, TabNet also support residual connection as well as batch normalization which assist the stability, robustness, and speed of learning.

Thus, this paper is organized as follows. Firstly, the related work on association football matches outcomes prediction using the available benchmark dataset. Then, a brief explanation and description on implementation of experiment which include dataset, experimental setup, modelling and scoring rules (performance metric). Next, the results of the experiment will be presented and discussed briefly, and finally conclude the paper with a summary of the whole experiment and its output.

RELATED WORK

The 2017 Soccer Prediction Challenge [2] has been participated by many researchers on how to develop a football outcomes prediction model from all over the world. The 216743 instances of past soccer matches consist of 18 seasons of football league between 2001/2002 to 2017/2018 across 35 countries. The main objective of the challenge is to develop football matches outcomes prediction model to predict 206 future match outcomes from 26 different leagues where the data used are limited to available data given. As the result, there are about 13 teams successfully participated. Table 1 shows the overall results of the 2017 Soccer Prediction Challenge [2] based on the ranking of the team in terms of average rank probability score (RPS) and accuracy as a performance metric where the lowest value of average RPS and highest value of accuracy indicates better prediction model.

TABLE 1. The results of the 2017 Soccer Prediction Challenge in term of accuracy and rank probability score.

Rank	Team	RPS _{avg}	Accuracy	Method
1	Team DBL*	0.2054	0.5194	[3]
2	Team OH	0.2063	0.5243	[12]
3	Team ACC	0.2083	0.5146	[1]
4	Team FK	0.2087	0.5338	[13]
5	Team DBL*	0.2149	0.5049	[3]
6	Team HEM	0.2177	0.4660	N/A
7	Team Priors	0.2255	0.4515	Prior information based on leagues
8	Team EB	0.2258	0.4854	N/A
9	Team Priors	0.2261	0.4515	Global prior of win, draw, lose
10	Team LJ	0.2313	0.4126	N/A
11	Team AT	0.3981	0.3883	N/A
12	Team LHE	0.4515	0.3398	N/A
13	Team EDS	0.4515	0.3592	N/A

[3] proposed a new prediction model using two machine learning techniques which are k -Nearest Neighbour (k -NN) and extreme gradient boosted tree (Xgboost). According to [3], k -NN is among the simplest and oldest supervised learning algorithms meanwhile describe the Xgboost as the latest update and powerful machine learning algorithm since the 17 out of 29 challenge winning solution of machine learning competition organized by Kaggle as well as every winning team in top-10 of KDD Cup 2015 are using Xgboost [10]. They also prepared two sets of learning feature methods which are the recency feature extraction method and rating feature method to be used with k -NN and Xgboost to predict a football match's outcome. As the result, k -NN with rating feature learning outperforms k -NN with recency feature extraction. In the 2017 Soccer Prediction Challenge, k -NN with rating feature learning remarkably have the lowest average value for RPS compared to other submitted models but were disqualified since they are the challenge's organizers as well as Xgboost based model successfully achieved rank 5 in the challenge.

[1] developed a new football prediction model called Dolores for the 2017 Soccer Prediction Challenge. Dolores is a product of two combining dynamic rating systems and a Hybrid Bayesian Networks (HBN) model. The dynamic rating system used to develop Dolores is originates from the work of [11] and was extend for further improvement by additional features represent team form. The HBN is developed to incorporates with the pi-rating as input to generate the predictive distribution result of home win, draw and away win. According to [2], Dolores which presented as "Team ACC" successfully achieved runner-up in the challenge which determines by a value of RPS. Despite achieved runner-up, the winner of the challenge [12] also adopted the pi-rating concept to model the latent variable for their prediction model.

[12] applied standard two machine learning algorithms which are Gradient Boosted Trees and Relational Dependency Networks (RDN)-Boost to predict the football matches results for football leagues around the world. The main contribution of their work is on the feature engineering process where their prediction model was developed based on the history of the matches played among football teams which are divided into relational and feature-based approaches. Six prediction models were generated namely baseline predictor, relational classification model with and without pi-rating, feature-based classification model, a feature-based classification that only considered pi-rating for prediction and feature based regression model. Six relevant features have been selected for developing the learning set for feature-based classification model and feature-based regression model include the feature on the historical strength of the team, the current form of the team, pi-rating, PageRank, the match importance and the league specification. The RPS has been choosing to evaluate all predictors and the feature-based classification model achieved the smallest value of RPS from season 2012 until season 2016. As the result, they decided to use the feature-based classification for participating in the challenge and successfully won the challenge by achieving the smallest value of RPS for predicting the 206 matches provided as a testing set.

Finally, [13] proposed two distinct approaches which are the hierarchical Poisson log-linear model (HPL) to predict goals scored in football matches and the Bradley-Terry model for predicting football match outcome. They divided features extraction into three categories namely team-specific, match-specific and season-specific. Then, several prediction models based on the Bradley-Terry model were developed determine by specification and extension in terms of features used namely baseline model (BL), constant strength (CS), linear with features (LF), time-varying coefficient (TVC) and additive features differences with time interaction (AFD) and each model used different method

to handling draw results whether ordinal or Davidson. In conclusion, the TVC model successfully outperform all models except HPL.

EXPERIMENT

The dataset, experimental setup, modelling which includes pi-rating system and TabNet and finally scoring rules will be described in this section. The raw dataset first will be calculated and optimized to pi-rating and then were learn and test using TabNet for football matches outcomes prediction. The subsection is as the following:

Dataset

The dataset was limited to 216,743 instances of football data consist of 18 seasons of football league between 2001/2002 to 2017/2018 across 35 countries as learning set extracted from [14] and 206 instances of football data as testing set extracted from [15] which have 9 features consist of season, league, data, home team, away team, home scored, away scored, goal difference and the matches results (Win, Draw, Loss). Figure 2 shows excerpt of learning and testing set as benchmark dataset to be used in this paper.

Sea	Lge	Date	HT	AT	HS	AS	GD	WDL
00-01	GER1	11/8/2000	Dortmund	Hansa Rostock	1	0	1 W	
00-01	GER1	12/8/2000	Bayern Munich	Hertha Berlin	4	1	3 W	
00-01	GER1	12/8/2000	Freiburg	VfB Stuttgart	4	0	4 W	
00-01	GER1	12/8/2000	Hamburger SV	Munich 1860	2	2	0 D	
00-01	GER1	12/8/2000	Kaiserslautern	Bochum	0	1	-1 L	
00-01	GER1	12/8/2000	Leverkusen	Wolfsburg	2	0	2 W	
00-01	GER1	12/8/2000	Werder Bremen	Energie Cottbus	3	1	2 W	
00-01	GER1	13/8/2000	Eintracht Frankfurt	Unterhaching	3	0	3 W	
00-01	GER1	13/8/2000	Schalke 04	FC Koln	2	1	1 W	
00-01	GER1	18/8/2000	Energie Cottbus	Dortmund	1	4	-3 L	
00-01	GER1	19/8/2000	Munich 1860	Werder Bremen	2	1	1 W	
00-01	GER1	19/8/2000	Bochum	Bayern Munich	0	3	-3 L	
00-01	GER1	19/8/2000	Hertha Berlin	Hamburger SV	4	0	4 W	

(a)

Sea	Lge	Date	HT	AT	HS	AS	GD	WDL
16-17	AUT1	1/4/2017	St Polten	Rapid Wien	1	1	0 D	
16-17	AUT1	1/4/2017	Austria Wien	Admira Wacker	0	2	-2 L	
16-17	AUT1	1/4/2017	Sturm Graz	Wolfsberger AC	4	0	4 W	
16-17	AUT1	1/4/2017	SV Mattersburg	SV Ried	2	1	1 W	
16-17	AUT1	2/4/2017	SCR Altach	RB Salzburg	0	5	-5 L	
16-17	BEL1	31/3/2017	Waregem	Anderlecht	1	2	-1 L	
16-17	BEL1	1/4/2017	Charleroi	Oostende	1	1	0 D	
16-17	BEL1	2/4/2017	Gent	Club Brugge	2	1	1 W	
16-17	CHE1	1/4/2017	FC Sion	FC Thun	2	1	1 W	
16-17	CHE1	1/4/2017	FC St Gallen	FC Basel	0	3	-3 L	
16-17	CHE1	2/4/2017	FC Luzern	FC Lugano	0	2	-2 L	
16-17	CHE1	2/4/2017	Young Boys	FC Vaduz	3	2	1 W	

(b)

FIGURE 2. (a) Excerpt of the learning set extracted from [14] (b) Excerpt of the testing set extracted from [15].

Experimental Setup

The experiment was conducted using RStudio Integrated Development Environment to implement the pi-rating system in R language and Spyder Integrated Development Environment for implementation of TabNet algorithm in Python. Each software runs on AMD Ryzen 5 3600 6-Core Processor CPU in Window 10 (64-bit) operating system with 16 GB of random-access memory (RAM) and GPU of RTX 2060 Super.

Modelling

The modeling process of the proposed football matches outcomes prediction model consist of two parts which are the implementation of pi-rating for football team rating system and the application of TabNet on the pi-rating system for prediction. Figure 3 shows the machine learning workflow for proposed football matches outcomes prediction.

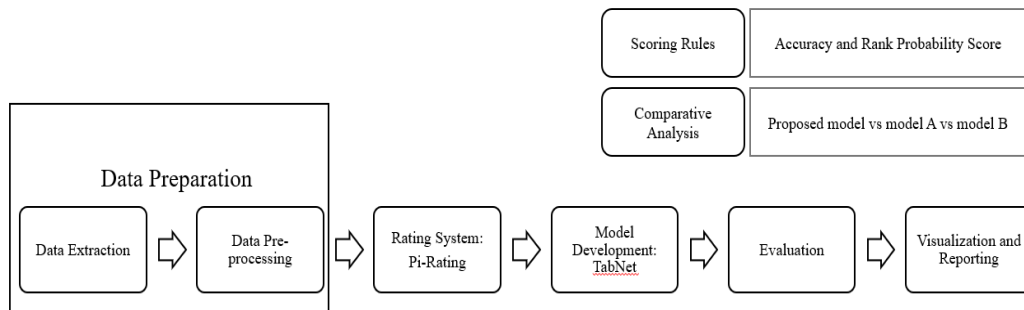


FIGURE 3. A machine learning workflow for proposed football matches outcomes prediction.

The football team rating system calls the pi-rating system was proposed by [11] considering three rules which are a home advantage, the importance of recent matches results and the fact that a win is more important for a football team compared to increase the number of goals difference. Thus, to define pi-rating, a football team that playing at home should consider their current home rating, their opponent's current away rating and the outcome of the match in order to estimate their new home rating and vice versa. The pi-rating has been used as a prominent football rating system during the 2017 Soccer Prediction Challenge [2] in the [1] and [12], where both of them successfully achieved 1st and 2nd places. The implementation of pi-rating has been conducted using the R package *piratings* developed by [16] where the R package assists to calculate and optimize the pi-ratings based on the work of [11]. Table 2 shows the description of variables from the learning and testing set as well as variables used to calculate football team pi-rating.

TABLE 2. The description of variables from the learning and testing set as well as variables used in pi-rating system

Variables	Details	Used in pi-rating
Season (Sea)	The season of football league edition	-
League (Lge)	The type of football leagues competition	-
Date	The date of the match	✓
Home Team (HT)	The team playing at home	✓
Away Team (AT)	The team playing at away	✓
Home Scored (HS)	The number of goals scored by home team	✓
Away Scored (AS)	The number of goals scored by away team	✓
Goal Difference (GD)	The difference between home team and away team	-
Matches Results (WDL)	The outcome of the match in term of win, draw and lose	-

TabNet

According to [8], Attentive Interpretable Tabular Learning Neural Networks or also known as TabNet is an extension of Deep Neural Networks (DNN) that has the advantage in data mining to learn and test tabular data. Generally, the DNN is popular for its functionality and specialty in images, text, and audio processing. It is known that ensemble methods-based tree models such as Xgboost [10], LightGBM [17], and CatBoost [18] are widely used and dominate most of the recent data science competition [19]. However, the work of [8] showed that TabNet incorporated both unsupervised and supervised learning and can challenge all both existing DNN and tree-based models. The implementation of TabNet is executed using python *pytorch-tabnet*, which supports data classification and regression.

Scoring Rules

The evaluation metrics or also known as scoring rules in this paper follow the same scoring rules that have been used in the 2017 Soccer Prediction Challenge [2]. The training set consists of 216743 instances meanwhile 206 instances as a testing set or football matches outcomes that should be predicted which a part of the challenge. So, accuracy and Rank Probability Scores (RPS) have been used as scoring rules to verify and validate our proposed models against previous competing prediction models such as [1], [3], [12] and [13]. The accuracy can be defined as in Equation 1,

$$Accuracy = \frac{\text{The total prediction results}}{\text{The total observed results}} \quad (1)$$

where the total number of correctly predicted results is divided by the total number of actual observed results. Meanwhile the rank probability score (RPS) can be defined as in Equation 2.

$$RPS = \frac{1}{r-1} \sum_{i=1}^{r-1} (p_i - e_i)^2 \quad (2)$$

where r is the number of potential outcomes, p_j is the forecasted probability of outcome j and pe_j is the actual probability of outcome j . The smaller value of score indicates better forecasts accuracy.

RESULTS AND DISCUSSION

The experiment is conducted using the same dataset that incorporates 216,743 instances of football data consist of 18 seasons of football league between 2001/2002 to 2017/2018 across 35 countries as a learning set and 206 instances of football data as testing set from 52 football league. The performance metrics or scoring rules were accessed via accuracy and RPS on those matches. The raw dataset was first transformed into pi-rating with consist the pi-rating at home ground and pi-rating at away ground. The rating difference also was measured between the home team and away team before the dataset was learning using TabNet. The current pi-rating for the home team and away team are then be inserted to 206 matches as well as the rating difference as testing set for prediction. Figure 4(a) show the excerpt of pi-rating generated from the raw dataset where $V1$ is the current pi-rating for the football team at home ground and $V2$ is the current pi-rating for the football team at away while Figure 4(b) is the excerpt of output for 206 instances of the testing set of football matches outcomes prediction using TabNet in term of accuracy and probability distribution (win, draw and lose).

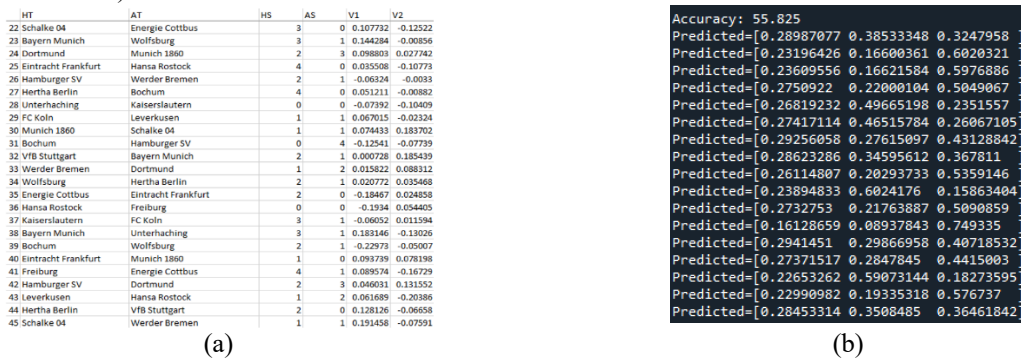


FIGURE 4. (a) Excerpt of pi-rating generated from the raw dataset where $V1$ is current pi-rating for the football team at home and $V2$ is current pi-rating for the football team at away (b) Excerpt of output for 206 instances of the testing set of football matches outcomes prediction using TabNet in term of accuracy and probability distribution (win, draw and lose).

Overall, the proposed models using TabNet based on pi-rating system successfully achieved 55.82% the highest in terms of performance accuracy and smallest in value of 0.1956 for rank probability score compared to other previous participants in the challenge. Table 3 shows the comparative analysis results of the proposed model against existing models in terms of accuracy and rank probability score.

TABLE 3. The comparative analysis results of the proposed model against existing models in term of accuracy and rank probability score.

Model	Accuracy (%)	Rank Probability Score
Berrar et al. (2019)	51.94	0.2054
Hubacek et al. (2019)	52.43	0.2063
Constantinou (2019)	51.46	0.2083
Tsokos et al (2019)	53.38	0.2087
Berrar et al. (2019)	50.49	0.2149
*Proposed Model	55.82	0.1956

It is observed that modelling football matches outcomes prediction may cause challenges due to the constraint of the limited accessible informative dataset. As a result, the integration of football domain knowledge, expert knowledge, and data engineering maybe the solution to produce good predictive football prediction models as well as the choices of techniques and algorithm whether following the statistical models, machine learning and probabilistic graphical models and rating systems.

CONCLUSION

To summarise, the 2017 Soccer Prediction Challenge successfully attract many researchers to gather and produce several fascinating football matches outcomes prediction models despite limited valuable informative dataset or in other words, basic football dataset that can be accessed online. It is also can be stated that concern in the football prediction model can be divided into three clusters: (1) football data, (2) approaches, techniques or algorithms used for modelling, and (3) the scoring rules (performance metric). For the challenge, it is observed that the football data and techniques are the main concerned meanwhile problem statement related to scoring rules to evaluate and validate football prediction models have been solved. In conclusion, this research was set to investigate the prediction performance for football matches outcomes based on a pi-rating system using TabNet a new Deep Neural Networks architecture for tabular data. The limited raw dataset was used to calculate the pi-rating system where some variables have been transformed to generate informative variables that can assist in football prediction. Then, the pi-rating will be used as a new dataset to predict football matches outcomes using TabNet in terms of the probability distribution of win, draw and lose. As the result, the proposed model successfully overcome all the top 5 models which participate in the challenge in terms of accuracy and RPS.

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